

ESAT CAMP MOCK TEST D

Physics

EC-PHY-D

PHYSICS

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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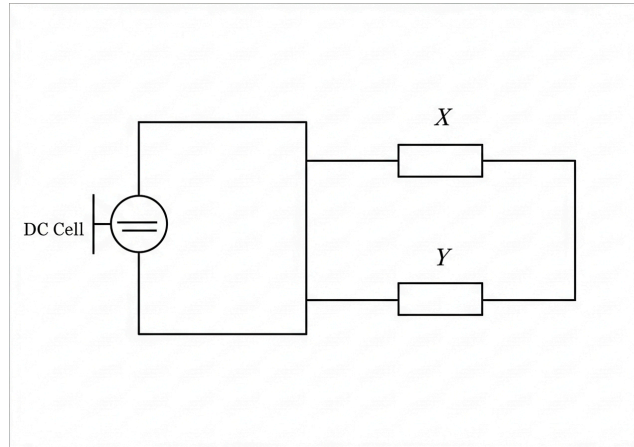
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Physics

- 1 Two wires, X and Y, are made from the same material. Wire Y has twice the length and half the radius of wire X.

The two wires are connected in parallel to a DC power supply.

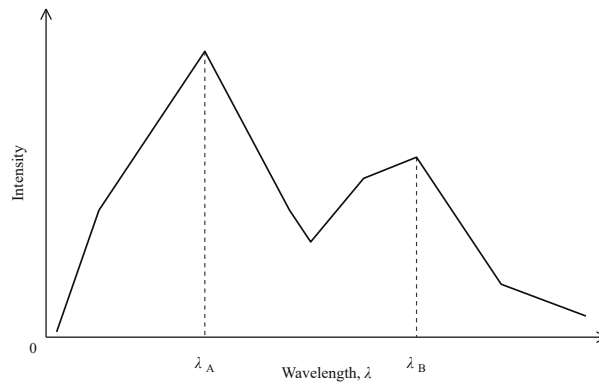
What is the ratio of the power dissipated by wire Y to the power dissipated by wire X, $\frac{P_Y}{P_X}$?



concept image

- A $\frac{1}{8}$
- B $\frac{1}{4}$
- C $\frac{1}{2}$
- D 2
- E 4
- F 8

- 2 The figure shows the intensity of an electromagnetic signal as a function of its wavelength, λ .



The signal has two distinct intensity peaks: a taller peak at wavelength λ_A and a shorter peak at a longer wavelength λ_B . This signal is then re-plotted to show its intensity as a function of frequency, f . Let f_A be the frequency corresponding to λ_A and f_B be the frequency corresponding to λ_B . Which statement correctly describes the new graph?

- A** The peak originally at λ_A is at a higher frequency than the peak from λ_B , and it is the taller peak.
- B** The peak originally at λ_A is at a lower frequency than the peak from λ_B , and it is the taller peak.
- C** The peak originally at λ_A is at a higher frequency than the peak from λ_B , but it is the shorter peak.
- D** The peak originally at λ_A is at a lower frequency than the peak from λ_B , and it is the shorter peak.
- E** The two peaks will appear at the same frequency on the new graph.
- F** The new frequencies are given by the relations $f_A = c\lambda_A$ and $f_B = c\lambda_B$.

- 3 A block of mass 4.0 kg rests on a rough plane inclined at an angle of 30° to the horizontal.

The coefficient of static friction between the block and the incline is $\mu = \frac{\sqrt{3}}{6}$.

A light, inextensible string attached to the block passes over a smooth, fixed pulley at the top of the incline. A second block of mass m is suspended vertically from the other end of the string, as shown in the diagram.

What is the complete range of values of m for which the system remains in static equilibrium?

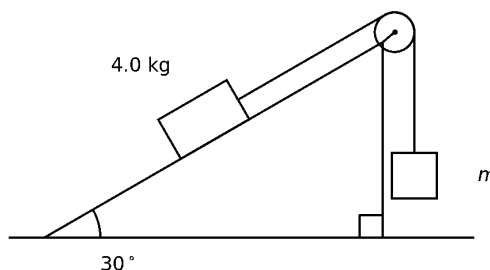


Diagram not to scale

- A $1.0 \text{ kg} \leq m \leq 3.0 \text{ kg}$
 - B $1.0 \text{ kg} \leq m \leq 2.0 \text{ kg}$
 - C $2.0 \text{ kg} \leq m \leq 3.0 \text{ kg}$
 - D $1.5 \text{ kg} \leq m \leq 2.5 \text{ kg}$
 - E $0.5 \text{ kg} \leq m \leq 3.5 \text{ kg}$
 - F $0.8 \text{ kg} \leq m \leq 3.2 \text{ kg}$
- 4 A fixed mass of an ideal gas is kept at a constant temperature inside a cylinder fitted with a piston. The piston is pulled outwards so that the volume of the gas increases at a constant rate.
- Which statement best describes the pressure of the gas?
- A It is constant.
 - B It is increasing at a constant rate.
 - C It is increasing at a decreasing rate.
 - D It is decreasing at a constant rate.
 - E It is decreasing at a decreasing rate.

- 5 Two solid spheres, P and Q, fall vertically through a fluid of density ρ . Sphere P has radius r and density 3ρ ; sphere Q has radius $2r$ and density 2ρ .

For this question, the upthrust equals the weight of displaced fluid, and the volume of a sphere is proportional to the cube of its radius. The resistive force on a sphere of actual radius s is $F = ksv$, where v is its speed and k is constant.

Both spheres reach terminal velocities v_P and v_Q . What is v_Q/v_P ?

- A 1
 - B 2
 - C $\frac{8}{3}$
 - D 4
 - E 8
- 6 An electric motor has an electrical power input of 160 W. The motor lifts a mass of 8.0 kg vertically at a constant speed of 1.5 m s^{-1} .

What is the efficiency of the motor? (Assume the acceleration due to gravity $g = 10 \text{ m s}^{-2}$.)

- A 7.5%
 - B 15%
 - C 25%
 - D 40%
 - E 50%
 - F 75%
- 7 A DC power supply of constant voltage is connected to a fixed resistive load of resistance $2R$ using connecting wires. The total resistance of the connecting wires is initially R . The connecting wires are replaced by new wires of the same material and cross-sectional area, but four times the length. What is the ratio of the power dissipated in the new wires to the power dissipated in the original wires?
- A 1 : 4
 - B 1 : 2
 - C 1 : 1
 - D 2 : 1
 - E 4 : 1

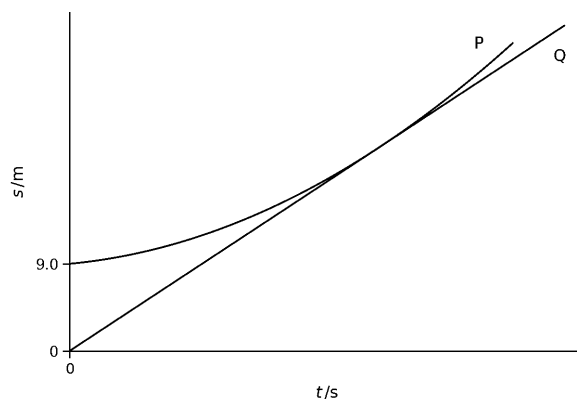
- 8 The displacement–time (s – t) graph shows the motion of two particles, P and Q, travelling in the same direction along a straight line.

Particle P starts at time $t = 0$ from displacement $s = 9.0$ m with an initial velocity of 1.0 m s^{-1} and moves with a constant acceleration of 2.0 m s^{-2} .

Particle Q starts at time $t = 0$ from displacement $s = 0$ and moves with a constant speed v .

On the graph, the straight line representing the motion of Q is tangent to the curve representing the motion of P.

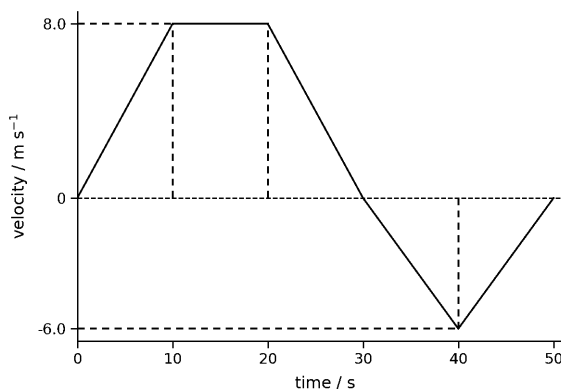
What is the value of v ?



- A 3.0 m s^{-1}
- B 5.0 m s^{-1}
- C 6.0 m s^{-1}
- D 7.0 m s^{-1}
- E 8.0 m s^{-1}
- F 9.0 m s^{-1}
- G 12.0 m s^{-1}
- H 13.0 m s^{-1}

- 9** A small parcel of liquid at the bottom of a container is heated by a source below it. The parcel expands and begins to rise, initiating a convection current. Which statement best explains the physical origin of the net upward force acting on this heated parcel?
- A** The temperature of the parcel increases, giving the molecules sufficient kinetic energy to overcome gravity and move upwards.
 - B** The density of the parcel decreases, causing the weight of the surrounding fluid it displaces to exceed the weight of the parcel itself.
 - C** The volume of the parcel increases, causing the internal pressure of the parcel to become significantly greater than the external pressure from the surrounding fluid.
 - D** The density of the parcel increases, resulting in a stronger reaction force from the container base that pushes the parcel upwards.
 - E** The mass of the parcel decreases due to the expansion, reducing the gravitational force acting on it.
- 10** The graph shows how the velocity of a test vehicle travelling along a straight horizontal track varies with time over a period of 50 s.
- What is the total distance travelled by the vehicle, its distance from its starting position at $t = 50$ s, and the magnitude of its average velocity over the 50 s period?

	Total distance travelled / m	Distance from starting position / m	Magnitude of average velocity / m s^{-1}
A	100	100	2.0
B	100	100	4.4
C	100	220	2.0
D	100	220	4.4
E	220	100	2.0
F	220	100	4.4
G	220	220	2.0
H	220	220	4.4



- 11** An electrical heater with a power output of 100 W is used to melt a large block of ice at 0 °C. The ice melts at a constant rate of 0.20 g s⁻¹.

What is the rate at which thermal energy is lost to the surroundings? (specific latent heat of fusion of ice = 3.5×10^5 J kg⁻¹)

- A** 30 W
- B** 40 W
- C** 70 W
- D** 100 W
- E** 170 W
- F** 70 kW

- 12** A block of mass 2.0 kg is placed on a rough slope inclined at 30° to the horizontal. It is connected by a light inextensible string passing over a smooth, frictionless pulley at the top of the slope to a hanging block of mass 3.0 kg, as shown in the diagram.

The coefficient of dynamic friction between the 2.0 kg block and the slope is $\frac{\sqrt{3}}{6}$.

When the system is released from rest, what is the magnitude of the acceleration of the blocks?

(Take the acceleration due to gravity to be $g = 10$ m s⁻².)

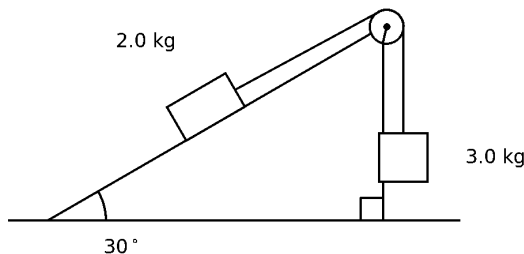


Diagram not to scale

- A** 1.0 m s⁻²
- B** 2.0 m s⁻²
- C** 3.0 m s⁻²
- D** 4.0 m s⁻²
- E** 5.0 m s⁻²
- F** 6.0 m s⁻²
- G** 7.5 m s⁻²
- H** 10 m s⁻²

- 13** A uniform, horizontal magnetic field has a magnetic flux density of 0.40 T and is directed from south to north.

A straight, horizontal conducting wire of length 0.50 m and mass 0.060 kg is oriented in an east-west direction, perpendicular to the magnetic field.

When a constant current is switched on in the wire, it experiences an initial vertical upward acceleration of 5.0 m s^{-2} .

What are the magnitude and direction of the current in the wire?

(gravitational field strength $g = 10 \text{ N kg}^{-1}$)

	magnitude of current / A	direction of current
A	1.5	from west to east
B	1.5	from east to west
C	3.0	from west to east
D	3.0	from east to west
E	4.5	from west to east
F	4.5	from east to west
G	6.0	from west to east
H	6.0	from east to west

- 14** A ray of light is directed horizontally towards a plane mirror M_1 which is inclined at an angle of 30° to the horizontal. The reflected ray strikes a second plane mirror M_2 and, after reflecting from M_2 , travels vertically upwards, as shown in the diagram.

Mirror M_1 is now rotated clockwise through an angle α , where $\alpha < 30^\circ$. The direction of the incident ray is unchanged.

In what direction and through what angle should mirror M_2 be rotated in order for the ray to still emerge vertically upwards after reflecting from mirror M_2 ?

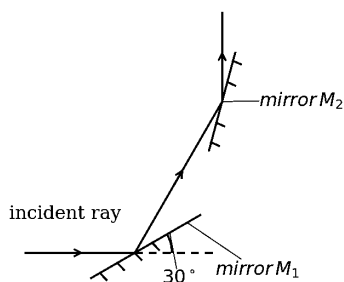


Diagram not to scale

- A** clockwise through an angle α
 - B** anticlockwise through an angle α
 - C** clockwise through an angle 2α
 - D** anticlockwise through an angle 2α
 - E** clockwise through an angle $\frac{\alpha}{2}$
 - F** anticlockwise through an angle $\frac{\alpha}{2}$
- 15** A ball is projected vertically upwards from the ground. It rises to a maximum height H and then falls back to the ground. Throughout the motion, the ball is subject to a force of air resistance that increases as the speed of the ball increases.

Let W_{up} be the magnitude of the work done by air resistance during the ascent.

Let W_{down} be the magnitude of the work done by air resistance during the descent.

Which of the following statements correctly compares W_{up} and W_{down} ?

- A** $W_{\text{up}} = W_{\text{down}}$, because the distance travelled is the same in both phases.
- B** $W_{\text{up}} > W_{\text{down}}$, because the speed at any given height is greater during the ascent.
- C** $W_{\text{up}} < W_{\text{down}}$, because the time taken for the descent is greater than for the ascent.
- D** $W_{\text{up}} < W_{\text{down}}$, because the ball accelerates due to gravity during the descent.

- 16** A periodic wave travels through a medium with a speed of 5.0 m/s. The period of the wave is 2.0 s.
- What is the number of complete wavelengths that fit into a distance of 120 m along the direction of propagation?
- A** 12
B 24
C 48
D 60
- 17** A variable resistor is connected to a power supply that provides a constant voltage. The resistance of the resistor decreases at a constant rate over a period of time, but remains positive.
- Which statement correctly describes the power dissipated by the resistor during this time?
- A** It is increasing at an increasing rate.
B It is increasing at a constant rate.
C It is increasing at a decreasing rate.
D It is not changing.
E It is decreasing at a constant rate.
F It is decreasing.
- 18** A small ball is released from rest from a height h above a horizontal floor.
- After each bounce, the maximum height reached by the ball is a constant fraction k of the height from which it fell immediately before that bounce.
- It is observed that the maximum height reached by the ball after its second bounce is $\frac{1}{4}h$.
- Assuming air resistance is negligible and the acceleration due to gravity is g , what is the total time that elapses from the moment the ball is released until it comes to rest on the floor?
- A** $(1 + \sqrt{2})\sqrt{\frac{2h}{g}}$
B $(2 + \sqrt{2})\sqrt{\frac{2h}{g}}$
C $(1 + 2\sqrt{2})\sqrt{\frac{2h}{g}}$
D $(3 + \sqrt{2})\sqrt{\frac{2h}{g}}$
E $(3 + 2\sqrt{2})\sqrt{\frac{2h}{g}}$
F $(4 + 2\sqrt{2})\sqrt{\frac{2h}{g}}$
G $3\sqrt{\frac{2h}{g}}$

- 19** A radioactive sample contains two different isotopes, P and Q, which decay independently into stable products.

At time $t = 0$:

- the number of nuclei of isotope P is 8 times the number of nuclei of isotope Q
- the half-life of isotope P, T_P , is 4 times the half-life of isotope Q, T_Q

Which row in the table gives the ratio of the activity of isotope P to the activity of isotope Q, $\frac{A_P}{A_Q}$, at time $t = 0$ and at time $t = T_P$?

	Ratio $\frac{A_P}{A_Q}$ at $t=0$	Ratio $\frac{A_P}{A_Q}$ at $t=T_P$
A	2	16
B	2	64
C	8	16
D	8	64
E	2	8
F	2	16
G	32	64
H	32	256

- 20** For an electrical circuit, the total power P delivered to it and the current I flowing through it are measured. A graph of the quantity $\frac{P}{I}$ on the y-axis against I on the x-axis is a straight line with a positive gradient and a positive y-intercept.

Which of the following could be the circuit?

- A** A single ideal resistor.
- B** Two ideal resistors in parallel.
- C** An ideal cell of constant emf, being charged.
- D** An ideal resistor in series with an ideal cell of constant emf, which is being charged.
- E** A discharging battery with internal resistance.
- F** A filament lamp.

- 21** A rectangular block of a semiconductor has a volume of 8.0 cm^3 . The electrical resistance is measured between each of the three pairs of opposite faces, giving values of 1.0Ω , 2.0Ω , and 4.0Ω .
- What is the resistivity of the semiconductor material?
- A** $8.0 \times 10^{-3} \Omega \text{ m}$
 - B** $4.0 \times 10^{-2} \Omega \text{ m}$
 - C** $4.0 \times 10^{-1} \Omega \text{ m}$
 - D** $2.0 \Omega \text{ m}$
 - E** $4.0 \Omega \text{ m}$
 - F** $1.0 \times 10^2 \Omega \text{ m}$
- 22** A resistor of constant resistance dissipates 100 W when the potential difference across it is V . The potential difference is reduced to $0.8V$.
- What is the new power dissipated by the resistor?
- A** 64 W
 - B** 80 W
 - C** 100 W
 - D** 125 W
- 23** A progressive wave of frequency 5 Hz travels from point P to point Q. The distance from P to Q is 6 m . An observer at Q detects a wave peak 0.10 s after an observer at P detects a peak.
- If the wavelength is known to be between 3 m and 5 m , what is the speed of the wave?
- A** 12 m/s
 - B** 20 m/s
 - C** 30 m/s
 - D** 60 m/s
- 24** Two spherical organisms, X and Y, have the same density and specific heat capacity. The radius of organism Y is three times the radius of organism X ($r_Y = 3r_X$). Both are exposed to a radiant heat source that provides uniform energy per unit surface area.
- Which option correctly identifies the ratio of total heat energy absorbed per unit time by Y compared to X, and the ratio of the initial rate of temperature rise of Y compared to X?
- A** Energy absorbed: $3 : 1$; Rate of temperature rise: $1 : 3$
 - B** Energy absorbed: $9 : 1$; Rate of temperature rise: $3 : 1$
 - C** Energy absorbed: $9 : 1$; Rate of temperature rise: $1 : 3$
 - D** Energy absorbed: $27 : 1$; Rate of temperature rise: $1 : 1$
 - E** Energy absorbed: $27 : 1$; Rate of temperature rise: $1 : 9$

- 25** A solid cuboid has a volume of 0.0040 m^3 and a uniform density of 2000 kg m^{-3} . The areas of its three distinct faces are in the ratio $1 : 2 : 8$.

What is the maximum pressure the cuboid can exert when it rests on a horizontal surface? (Use gravitational field strength $g = 10 \text{ N kg}^{-1}$)

- A** 100 Pa
- B** 800 Pa
- C** 1000 Pa
- D** 4000 Pa
- E** 8000 Pa

- 26** A sealed, fixed-volume vessel containing an ideal gas is placed in a large room of constant temperature and allowed to cool. The rate of heat loss from the gas is proportional to the temperature difference between the gas and the room. At time $t = 0$, the pressure of the gas is 240 kPa. At $t = 20$ minutes, the pressure is 120 kPa. The half-life of the excess pressure is 10 minutes. The excess pressure is the difference between the gas pressure and its final equilibrium pressure.

What are the final equilibrium pressure of the gas and the pressure at $t = 30$ minutes?

- A** $P_f = 80 \text{ kPa}$; $P(30) = 90 \text{ kPa}$
- B** $P_f = 80 \text{ kPa}$; $P(30) = 120 \text{ kPa}$
- C** $P_f = 120 \text{ kPa}$; $P(30) = 135 \text{ kPa}$
- D** $P_f = 80 \text{ kPa}$; $P(30) = 100 \text{ kPa}$
- E** $P_f = 0 \text{ kPa}$; $P(30) = 60 \text{ kPa}$

- 27** The magnitude of the gravitational force, F , between two point masses with masses M_1 and M_2 is measured. The distance between their centres is r .

Which of the following proportionalities correctly relates these quantities?

- A** $F \propto \frac{M_1 + M_2}{r^2}$
- B** $F \propto \frac{M_1 M_2}{r}$
- C** $F \propto \frac{M_1}{r^2}$
- D** $F \propto (M_1 M_2) r^2$
- E** $F \propto \frac{M_1 M_2}{r^2}$